

Mathematics

A Cultural Heritage

Lecture 2

Numbers – Part 5

Operations on Numbers

Operations on Numbers



"Three" "plus" "Two"
"equals" "Five"

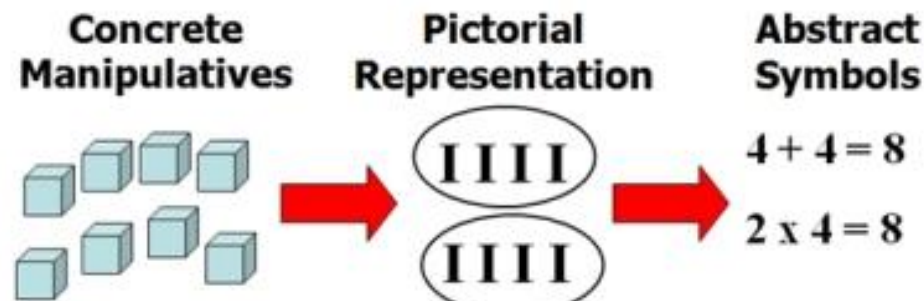
Goals of Mathematics

- **Particular cases** to **general situation**



$$3 + 2 = 5$$

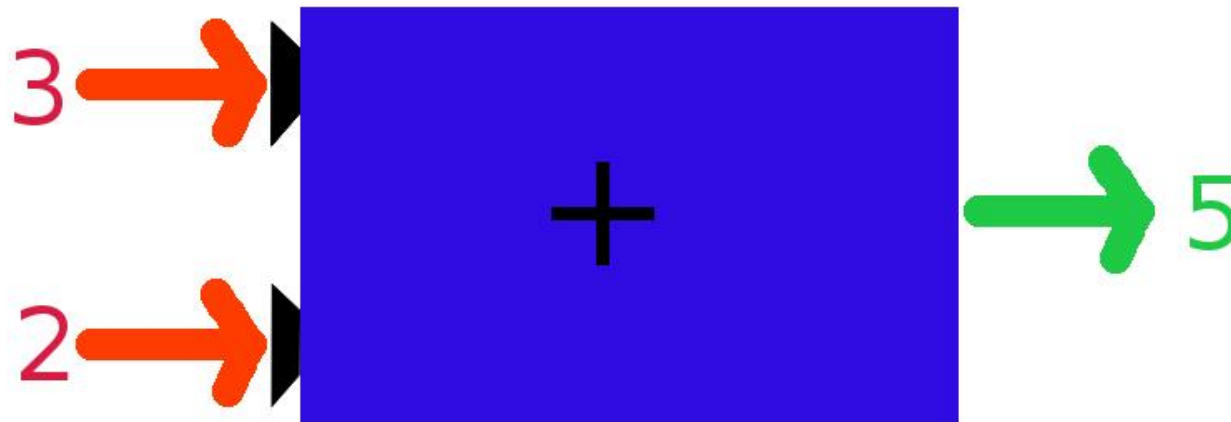
- **Concrete** to **abstract**



Source: learningbridges.com

Functions

1. Functions take **inputs** and give **outputs**



2. One or many inputs.
3. Inputs uniquely determine output.

Logarithm

- John Napier:
 - Napier's Bones
 - Logarithm Tables

Gr.	9					
min	Sinus	Logarithmi	Differentia	logarithmi	Sinus	
0	1564345	18551174	18427293	123881	9876883	60
1	1567218	18532826	18408484	124342	9876427	59
2	1570091	18514511	18389707	124804	9875971	58
3	1572964	18496231	18370964	125267	9875514	57
4	1575837	18477984	18352253	125731	9875056	56
5	1578709	18459772	18333576	126196	9874597	55
6	1581581	18441594	18314933	126661	9874137	54
7	1584453	18423451	18296324	127127	9873677	53
8	1587325	18405341	18277747	127594	9873216	52
9	1590197	18387265	18259203	128062	9872754	51
10	1593069	18369223	18240692	128531	9872291	50
11	1595941	18351214	18222213	129001	9871827	49
12	1598812	18333237	18203765	129472	9871362	48
13	1601684	18315294	18185351	129943	9870897	47
14	1604555	18297384	18166969	130415	9870431	46
15	1607426	18279507	18148619	130888	9869964	45



Napier (1550-1617)
[source: wikipedia]



Napier's bones
[source: wikipedia]

Logarithm and Exponential

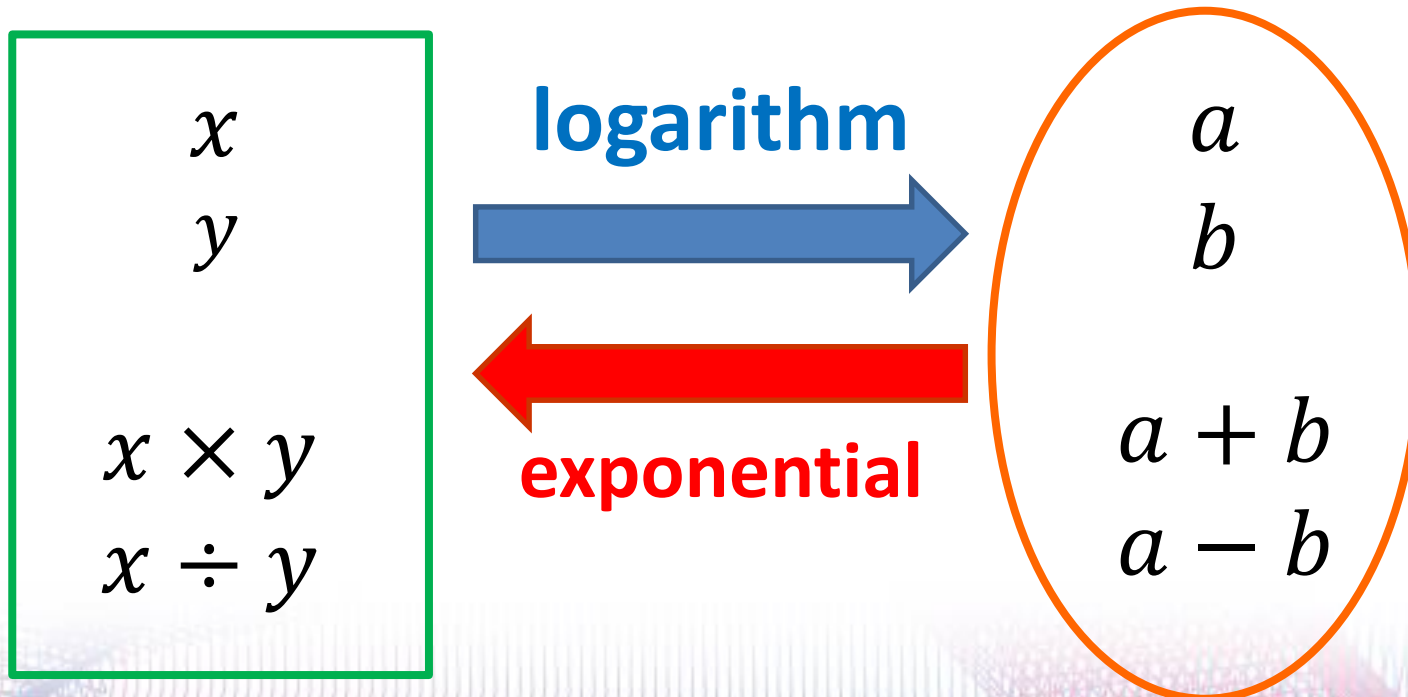
- Real number x .
- If $10^b = x$, b is logarithm of x .

$$\log(x) = b$$

- Log can be positive, negative, zero.

Exponential and Logarithm Rules

- $10^a \times 10^b = 10^{a+b}$
- $10^a \div 10^b = 10^{a-b}$
- Let $x = 10^a, y = 10^b$. Then:



Point of Logarithms

- Why were these useful?
- Try (by hand, on paper) to do the following:

$$23.45076109357287 \times 50329.3428705632310457$$

$$1.37015694229097584 + 4.70182126003053114$$

Abstraction and Mathematical Worlds

- Idea: Move question to an easier setting
- Examples:
 - Numbers/Correspondence/Tallying
 - Addition
 - Logarithm/Exponential
 - Mazes/Graphs (weekly reading)

Reading Assignment

1. Video: **The Story of 1** (BBC, 2005)
<https://www.youtube.com/watch?v=KoSWtEWTvWE>
2. Book excerpt: **The language of number**, in *The Story of Numbers: How Mathematics Has Shaped Civilization*, John McLeish, Fawcett Columbine Book, 1991, p.7-20.
3. Webpage: **A Brief History of Zero**, Kristen McQuillin, July 1997 (revised January 2004)
<http://www.mediatinker.com/blog/archives/008821.html>

Summary

- More and more general numbers throughout history
- Not always accepted right away
- Increasingly became part of mathematical theory
 - More generality
 - Richer mathematics
 - Better understanding of original concepts
 - Applications

Further Optional Readings

On physiological basis of human's ability in mathematics

- Our Brains Have a Map for Numbers,
<http://www.scientificamerican.com/article/our-brains-have-a-map-for-numbers/>
- Where Mathematics Comes From,
http://en.wikipedia.org/wiki/Where_Mathematics_Comes_From

References

1. Pika, Simone; Nicoladis, Elena; and Marentette, Paula (January 2009). "How to Order a Beer: Cultural Differences in the Use of Conventional Gestures for Numbers". *Journal of Cross-Cultural Psychology* 40 (1): 70–80.
2. Anne Rooney, *The Story of Mathematics*, Arcturus, 2009.
3. *The Story of 1* (Video program), BBC, 2005.